

Agenda

1. Announcements

- 11.2 and 11.3 HW due today 
- Discussion Quiz 7 tomorrow over
- Midterm 2 is on Tuesday April 11 at 7:30pm
 - Check for conflicts and sign up for the makeup if needed
 - Sign up to take exam at the testing center if you have testing accommodations

2. Section 11.5 - Alternating Series

3.) Section 11.6 - Ratio and Root test

Activity: Does $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\ln(n+4)}$ converge? $\overbrace{a_n}^1$

check

decreasing
positive

$\lim_{n \rightarrow \infty} a_n = 0$

a_n decreasing

$$\sum (-1)^{n-1} \left(\frac{1}{\ln(n+4)} \right)$$

$$n+4 \leq (n+1)+4 = n+5$$

$$\ln(n+5) \geq \ln(n+4) \quad y = \ln x \text{ is increasing} \quad \checkmark$$

$$a_n \geq 0 \quad \ln x \geq 0 \text{ for all } x \geq 1 \quad \checkmark$$

$$\ln(n+4) \geq 0 \text{ because } n+4 \geq 1+4 = 5 > 1$$

$$\lim_{n \rightarrow \infty} \frac{1}{\ln(n+4)} = \frac{1}{\infty} = 0 \quad \checkmark$$

Therefore $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\ln(n+4)}$ converges by the Alt series test.

Definition: Conditionally Convergent

when series converges but $\sum |a_n|$ diverges.

$$\textcircled{\text{ex}} \quad \sum \frac{(-1)^{n-1}}{n} \quad \left| \frac{1}{n} \right| \quad \frac{1}{n^p}$$

$p \leq 1$, divergent

possible types of questions:

- 1) determine if (series) converges or diverges
- 2) determine if (series) absolutely converges, conditionally converges, or diverges.

only possible if negative terms show up

Estimating the Sum Using Alternating Series Test

Remainder Estimation

Example: Estimate the sum of $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$ using S_{10}

Ratio Test

Let $\sum a_n$ be a series.

- 1.) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$, then $\sum a_n$ absolutely converges
- 2.) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1$, then $\sum a_n$ diverges includes ∞
- 3.) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$, then the ratio test is inconclusive.

This test works well for series with a combination of exponentials & factorials

Useful skill:
Simplifying factorials.

Example: Apply the ratio test to $\sum_{n=1}^{\infty} \frac{2^n}{n!}$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{2^{n+1}}{(n+1)!}}{\frac{2^n}{n!}} \right|$$

$$a_n = \frac{2^n}{n!}$$

$$a_{n+1} = \frac{2^{n+1}}{(n+1)!}$$

$$= \lim_{n \rightarrow \infty} \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n}$$

$$= \lim_{n \rightarrow \infty} \frac{\cancel{2^{n+1}}}{2^n} \cdot \frac{\cancel{n!}}{(n+1)!} = \lim_{n \rightarrow \infty} \frac{2}{n+1} = 0 < 1$$

$\therefore \sum_{n=1}^{\infty} \frac{2^n}{n!}$ converges by the ratio test.

Activity Apply the ratio test to $\sum_{n=1}^{\infty} \frac{n^4}{(-5)^n}$

$$= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$a_{n+1} = \frac{(n+1)^4}{(-5)^{n+1}}$$

$$= \lim_{n \rightarrow \infty} \left| \frac{(n+1)^4}{(-5)^{n+1}} \cdot \frac{(-5)^n}{n^4} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{(n+1)^4}{n^4} \cdot \frac{(-5)^n}{(-5)^{n+1}} \right| = \lim_{n \rightarrow \infty} 2 \left(\frac{n+1}{n} \right)^4 = 2 > 1$$

$\therefore$ diverges by the ratio test

Example/Activity: Apply the ratio test to $\sum_{n=3}^{\infty} \frac{ne^{4n}}{(n-2)!(2n)!}$

Root Test

Suppose $\sum a_n$ is a series

1.)

2.)

3.)

Example: Apply the root test to $\sum_{n=1}^{\infty} \left(\frac{3n^2 - 2}{n^2 + 3} \right)^{2n}$

Activity: Apply the root test to $\sum_{n=1}^{\infty} \frac{(-1)^n}{(\ln n)^n}$